

Simplifying Facial Animation using Deep Learning based Phoneme Recognition

Vladimir Ivanov
vivanov@blizzard.com
Blizzard Entertainment
USA

Parag Havaldar
phavaldar@blizzard.com
Blizzard Entertainment
USA

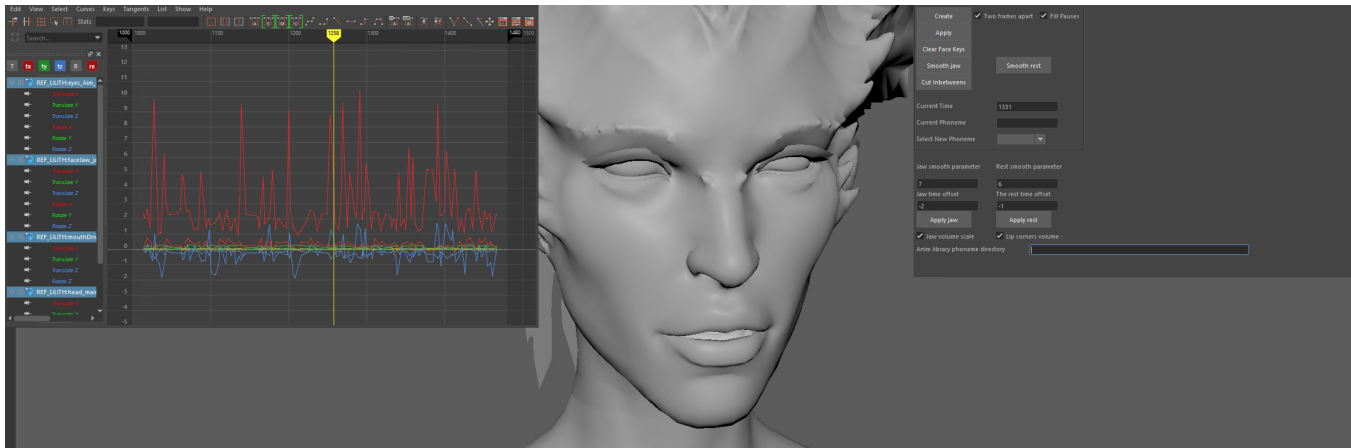


Figure 1: Facial Animation workflow and setup in Maya

ABSTRACT

We present a machine learning framework for facial animation that is simple, easy to implement and integrates well into an artist friendly workflow. The framework employs a pre-trained deep learning model used for phoneme extraction. Each phoneme is mapped to artist defined face shapes resulting in sparse key frames generating quick and synchronized looking dialogue animations, which can optionally be enhanced using familiar artist friendly workflows.

KEYWORDS

Facial Animation, Machine Learning

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1 OVERVIEW

Facial animation is a time consuming process requiring animators to animate high level phoneme poses corresponding to particular sounds along with low level controls. Getting the synchronization correct is not only hard but also not easy to iterate with. Our method simplifies this process by delegating these tedious timing tasks to a pretrained neural network which extracts synchronized phonemes. Each phoneme is mapped to an animator guided facial shape. For close up shots, our experimentation shows twenty shapes are needed for high fidelity, whereas twelve shapes suffice for background characters. In our system, the animators create the facial shapes and iterate over the mapping. The output of the machine learning process generates a firm starting point of sparse curves for all high level/low level controls that the animator used to define the phoneme poses. The framework also provides basic control over the curves such as volume adjustment of the jaw curves, time-offsetting the controls curves to account for the fact that muscles move earlier than the sound emerges. These sparse key setups can be further intuitively enhanced using regular workflows.

2 RELATED WORK

Many recent applications of machine learning for 3d facial animation can be divided into those that use video (or visual) inputs and that uses audio waveforms as input. In [Karras et al. 2017] the authors deep learning approach maps audio waveforms to fully animated and tracked 3D vertex coordinates of the whole face. While the results are good, the training is actor specific, and the output animated 3d vertex coordinates are very hard to edit with animator augmented workflows. In [Siarohin et al. 2020] the neural network

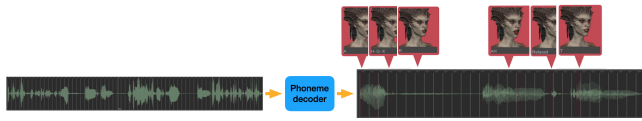


Figure 2: Model decodes phonemes from the input audio.

augments input image to match the driving video. Other related work includes [Navarro et al. 2021], where video is used as input to regress a set of FACS poses for each frame. The intent of this approach was to address realtime performance resulting in dense keys at every frame, which is not as friendly to an "animator in the loop" workflows.

3 METHOD

We use audio waveform as input and produce sparse rather than dense keys in the output animation. In our approach, we key the rig controls rather than working directly with the mesh, which incorporates our framework into an animator friendly editable workflow. Additionally we approach the solution hierarchically separating the jaw animation from the rest of the lip controls. Our process starts with the input audio, which is then parsed by a neural network akin to the one introduced in [Xu et al. 2022] to produce phonemes and other parts of speech synchronized to times on frames. Each phoneme that is defined in the training network is mapped to a specific face pose on the character. The exact shape of the pose is influenced by the animator using rig controls. As a result, the output fits seamlessly into animator's workflow since the framework provides sparse keys onto the rig controls. In this workflow, the mapping of phonemes to facial shapes can be iterated on by the animator, reassigned to other shapes, adjusted in time etc.

While the neural network guarantees synchronized output to hit parts of speech on the timeline, our post processing allows animators to control the visual output quality at such instances in time. In particular, we separate the jaw from the rest of the controls. The jaw amplitude is adjusted to account for the volume of the input audio. Since anatomically the muscles move before the sound appears, we apply a time offset to the controls so that they start moving before sound is heard generating perceived natural lip movement for speech. Another common option available to the animator is to skip phonemes that lie close (within one frame) from the previous phoneme to accentuate the dominant shape. The training framework is phoneme based (and not language based), and therefore lends itself well to other various spoken languages. This is illustrated well with the accompanying video. We additionally have added a "jaw swing" capability to the animation. The larger the amplitude of the swing, the more cartoonish the animation looks. Even for realistic animations it helps animation workflows to make facial dynamics more natural.

For further steps, we would like to explore extracting emotion in the speech to augment the lip-sync with suggestive movement around the eyes and brow, for example, animating the eyelid tighter, inner/outer brow and furrows to better express the emotions of the speaker. Since our process sets keys on frames only during the times where phonemes were detected, the long pauses are left

without proper attention. We are looking into filling in the gaps where only sighs and breathing are present.

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